

Investigation of Electrical Properties of LDPE/ZnO Nanocomposite Dielectrics

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ABSTRACT

In order to investigate the effect of inorganic nanoparticle fillers on the electrical properties of low density polyethylene (LDPE), LDPE/ZnO nanocomposites were prepared with a variety of filler loadings by melt blending method and the corresponding electrical properties are investigated. Experiments show that, the trap density is increased by 3-5 times in the nanocomposite as compared with LDPE. The conduction current of the nanocomposite is decreased to 0.5 – 0.25 of the value of LDPE both at low and high electric fields. The electrical breakdown strength is increased at low filler contents and decreased at high contents (>1 wt%). The electrical treeing life time is elongated 50 times at most in the nanocomposites. Both homo- and hetero-space charges are remarkably suppressed in the nanocomposite with 0.5 and 7 wt% contents. The study indicates that the improvements on the electrical properties, especially the charge transport and space charge suppression in the nanocomposites are closely related to the significant increase of deep trap density. The effect of UV light absorption and electric field homogenization by ZnO can also contribute to the enhancement of electrical breakdown strength and elongation of electrical treeing life.

Index Terms — Nanodielectrics, LDPE/ZnO, electrical property, space charge.

1 INTRODUCTION

POLYMERS have long been widely used as electrical insulations and dielectric materials for their excellent electrical, mechanical and thermal properties and easy processing characteristics [1]. However, improved properties are urgently expected with the growing pursuit of electrical equipments and electronic devices for high power density, high reliability, high durability in harsh conditions, etc. [2, 3]. It has already been proved that the additions of inorganic nanoparticles can enhance the mechanical and thermal properties of polymers [4, 5]. Recently, nanodielectrics (short for polymer based nanocomposite dielectrics) consisting of nanosized particle additives have drawn enormous attentions due to their dramatically improved electrical properties [6, 7]. Amongst the possible choices, one of the options is a polymer nanocomposite system with zinc oxide (ZnO) as the filler material, considering the fact that ZnO is an important wide band-gap semiconducting material with good UV absorption capabilities [8]. Such nanocomposites are being increasingly investigated at filler loadings higher than 10 wt% for nonlinear conductivity applications [8-11]. However, such a high filler loading can easily lead to the decrease of electrical insulation properties.

Hence it would be of great interest to investigate the electrical properties of these nanocomposites with lower loadings of ZnO fillers. Polyethylene (PE) is frequently adopted as the polymer matrix because of its wide use as electrical insulation materials for power cables, enormous available experimental data and the relatively simple structure. It has been reported recently that the addition of nanosized MgO particles to LDPE (low density polyethylene) improved the breakdown strength and volume resistivity, suppressed space charge accumulation at high electric field while exacerbated it at low electric field [12]. In this paper, we present a systematic and comparative study of the effect of lower filler content (<10 wt%) on the charge trap density, conduction current, electrical breakdown strength, electrical treeing life, space charge suppression properties of LDPE/ZnO nanocomposites.

2 MATERIALS

The nanocomposites were prepared by melt blending method. The ZnO nanoparticles (untreated) have an average particle size of about 50 nm prepared by homogeneous precipitation method [13]. LDPE is provided by Daqing petrochemical Company marked as 18D. LDPE and ZnO nanoparticles were melt-blended in a Haake batch mixer at 423 K and the nanocomposites with ZnO concentrations of 0.1, 0.5, 1, 3, 5, 7 and 10 wt% were produced. The obtained

nanocomposites were pressing molded at 393 K under a pressure of 10 MPa to obtain films with a thickness of about 100-200 μm . All samples were evaporated with Al electrodes on both sides with a diameter of 25 mm.

3 EXPERIMENTAL PROCEDURES

3.1 SCANNING ELECTRONIC MICROSCOPE IMAGING

The sample section surface, fractured in liquid nitrogen, was observed by scanning electron microscopy (SEM, Philips Quanta 200) operated at 20 kV in order to observe the dispersion of nanofillers.

3.2 ISOTHERMAL DISCHARGE CURRENT (IDC) MEASUREMENT

The trap level distributions in LDPE and LDPE/ZnO nanocomposites are determined from the modified isothermal discharge current method, which has been presented in detail in the published papers by the authors [14]. We only show the experimental process here. The specimen was firstly polarized at 323 K with a negative dc electric field of 30 kV/mm for 40min. Then at a constant temperature 323 K, the discharge current generated by the detrapped electrons was measured immediately after the specimen was short-circuited.

3.3 CONDUCTION CURRENT MEASUREMENT

Conduction current was measured by a picoammeter EST122 and a data acquisition system. The negative dc electric field was increased by 5 kV/mm from 5 kV/mm to 50 kV/mm and the steady state current values are recorded for each electric field. The measurement was performed under room temperature in a drying oven.

3.4 ELECTRICAL BREAKDOWN STRENGTH TESTING

The electrical breakdown strength was evaluated using CS2674C electrical strength test instrument. The voltage was raised from zero up to the level of breakdown at a rate of 1 kV/s. When breakdown occurred, the voltage supply was interrupted automatically and the breakdown voltage was recorded. For seven groups of specimens (0, 0.1, 0.5, 1, 3, 5 and 7 wt%), each one consisting of ten samples with the same thickness, the electrical breakdown strength was calculated and the average value and the deviation were obtained.

3.5 ELECTRICAL TREEING LIFE TESTING

The samples were cut to cube slabs 3cm for each side. Needle electrode (with a diameter of 1mm and radius of 3 μm) were hot-inserted into the sample as the high voltage electrode and the plate semiconducting material was hot-pressed to bind to the sample as the ground electrode. The distance between the needle electrode and the plate electrode is about 3 ± 0.5 mm. 10 kV ac voltage was applied to five samples simultaneously and the mean treeing breakdown time was recorded as the electrical treeing life of this kind of sample.

3.6 SPACE CHARGE DISTRIBUTION MEASUREMENT

Space charge distribution measurement was performed based on pulsed electro-acoustic (PEA) method under dc 50 kV/mm at room temperature. The principle and application of PEA method has been widely discussed in previous publications [1]. The instrument was marked as FIV-PEANUTS provided by Japan Five Lab Co., Ltd. The space resolution and charge resolution of the instrument is about 10 μm and 0.16 C/m³ respectively.

4 RESULTS AND DISCUSSION

4.1 NANOPARTICLE DISPERSION CHARACTERIZATION

Dispersion of the ZnO nanoparticles in LDPE/ZnO 5 wt% is presented by SEM image in Figure 1. As showed by the SEM section image, the ZnO nanoparticles are uniformly distributed in LDPE. The particle size is in the range 80-200 nm, with most particle size around 100 nm. A few large agglomerates can be observed.

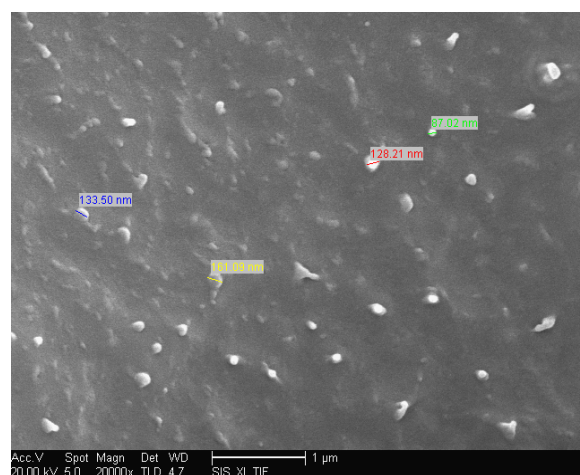


Figure 1 Sectional morphology graph of LDPE/ZnO 5%.

4.2 TRAP LEVEL DISTRIBUTIONS

Figure 2 shows the trap level distributions of LDPE and LDPE/ZnO nanocomposites with different filler loadings. The results show that the trap levels are both in the range 0.8-1.05 eV with a peak density around 0.94 eV, which can be regarded as deep trap levels in LDPE [15]. This is consistent very well with the results obtained by M. Ieda, who assigned the deep trap level in LDPE to be 0.92 eV [16]. The comparison of the trap level density is LDPE < LDPE/ZnO 3 wt% < LDPE/ZnO 5 wt% < LDPE/ZnO 0.5 wt%. The peak trap level density in LDPE and LDPE/ZnO 0.5 wt% nanocomposite are 8×10^{20} and $3.8 \times 10^{21}/(\text{eV}\cdot\text{m}^3)$ respectively, which are of the same order of the existing results (10^{16} - 10^{22} m³.eV) [14, 16-18].

As can be seen clearly, large amounts of deep traps are introduced by ZnO nanoparticle fillers. Traps have a profound effect on the charge carrier transport process and are closely related to most of the electrical properties of the dielectric materials in many aspects as will be showed later.

Traps are essentially localized states, where electrons (or holes) are immobilized to a certain degree. Trapping states in polymers are mainly due to chemical impurities and structure inhomogeneity. LDPE is a semi-crystalline polymeric material with most of the deep trapping states mainly located in the interface areas and amorphous regions [19], which has been demonstrated by Ieda via TSC and TL experiments [16]. In nanocomposite dielectrics, there are large amounts of interface areas between the nanoparticles and the polymer matrix due to the large specific surface area of the nanoparticles. It has already been recognized that interface areas play an important role in determine the electrical behavior of nanodielectrics [6, 20, 21]. As indicated by Tanaka and Nelson, the polymer molecules in the interface areas are partly immobilized due to their interaction with the nanoparticles and the space steric hindrance of the nanoparticles. The change in the local structures (free volume, chain arrangement, mobility, etc.) of the interface polymers may increase the density and depth of the charge carrier trapping states [22-24]. The fact that nanoparticle fillers can lead to a change in the polymer morphology structure and an increase in the trap level density has been confirmed by numerous experimental results [14, 25, 26].

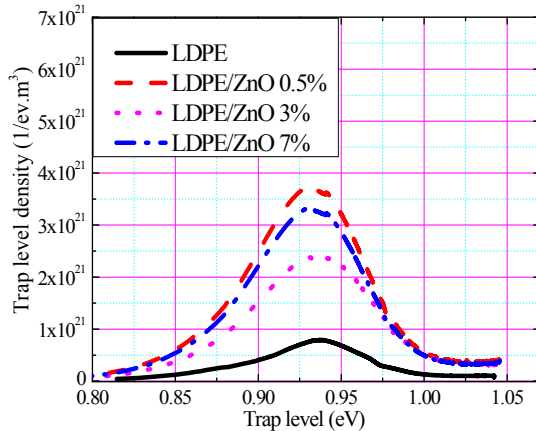


Figure 2 Trap level distribution LDPE and LDPE/ZnO nanocomposites.

4.3 CONDUCTION CURRENT

Figure 3 shows the conduction current of LDPE and LDPE/ZnO nanocomposites with different filler loadings. It can be clearly found that the conduction current in the LDPE/ZnO nanocomposites is much smaller (about 0.5-0.25) than that in LDPE for both low and high electric fields. The $\log J$ - $\log E$ plot exhibits ohmic characteristic at low fields (≤ 10 kV/mm) and space charge limited current (SCLC) characteristic at high fields (>10 kV/mm). The slope of the $\log J$ - $\log E$ in the SCLC region is larger than 2. This is the characteristic of SCLC with distributed trap levels [27-29]. The current in the presence of traps can be described as follows,

$$j = j_0 \cdot \frac{N_C}{N_t} e^{-E_t/kT} \quad (1)$$

j_0 is the ohmic without charge traps. N_C and N_t are the states density of the conduction band and trap levels respectively.

E_t is the trap depth. k is Boltzmann constant and T is the temperature. As it shows clearly, either the increase in the trap depth or its density can lead to a decrease in the current density. We have known that the trap level density of the nanocomposites is about 3-5 times the value of LDPE and the trap depth is generally unchanged as showed in Figure 2. So the current decrease in the nanocomposites can be attributed to the increase in trap level density, which can lead to a significant decrease in carrier mobility or free charge density equivalently.

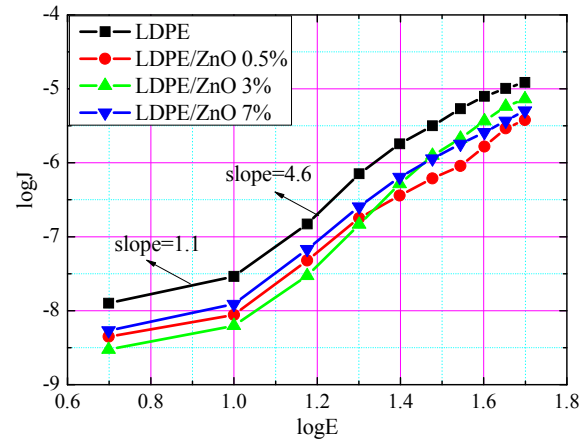


Figure 3 Conduction current characteristic of LDPE/ZnO nanocomposite.

4.4 ELECTRICAL BREAKDOWN STRENGTH

It has been recognized that short time electrical breakdown is mainly an electronic process [17, 30, 31]. It depends mainly on two factors, i.e. the density of free electrons and the energy they can gain. Electrons can be from electrode injection or impurity emission. The energy of the free electrons can be increased by acceleration in the electric field. If the free electrons gather enough energy, then they can cause the ionization of macromolecules and further initiate the electron avalanche. This will erode the materials in the form of partial discharge and leads to the formation of conducting channels and the final breakdown [32]. For polymers, since the free path length of electrons is short and the injected electrons from the electrode are quickly trapped after a few scatterings and spend most time in the traps. It is highly improbable that electron avalanches may occur in polymers. So it is commonly believed that large voids, or low density regions, in which electrons can gain enough energy for impact ionization, are formed at the stage of preparation for the breakdown. Large voids and micro-voids are formed during manufacture by evaporation of decomposition products in various chemical reactions induced by hot electrons and electron and hole recombination, or by impurities and additives decomposition or by migration, or by the interface defect at the large spherulite boundary, electromechanical stress induced crack, etc. [32]. In this view, any measures that can decrease the free electron density, carrier mobility, macroscopic void defects, will contribute to the enhancement of electrical breakdown strength of polymers. However, Electron emission by impurities or shallow traps at high electric field and thermal activation will increase the free electron density and then leading to the decrease of the electrical breakdown strength.

ZnO nanoparticle fillers can affect the electrical breakdown strength in the following aspects.

- Nanoparticles can serve as heterogeneous nucleation agent, which can accelerate the crystalline speed and prohibit the formation of large spherulites [33, 34]. This will decrease the formation of macroscopic boundary voids;
- Nanoparticles can become scattering sources for electrons [35];
- The interface domain can introduce large amounts of charge traps, which will dramatically decrease the carrier mobility [36], as has been discussed earlier in part 4.2 and 4.3;
- Nanoparticles can hinder the polymer from the erosion of partial discharge [22];
- Nanoparticle aggregates can induce large void defects around them;
- The large permittivity of nanoparticles can distort the electrical field around them, especially when the aggregates are large;
- As the content increases, the distance between nanoparticles decreases, which can lead to the formation of conductivity pathways and increase the tunneling current between nanoparticles.

As can be clearly seen from Figure 4, the ac electrical breakdown strength of the pure LDPE is 90 kV/mm and the higher value is obtained for ZnO 0.1, 0.5 and 1 wt% contents. With the ZnO contents increased from 0.1 to 7 wt%, the electrical breakdown strength decreased from 100 to 70 kV/mm. This may be due to that, at low filler loadings the decisive factors are (a)-(d); but at high filler loadings, (e)-(g) becomes dominant.

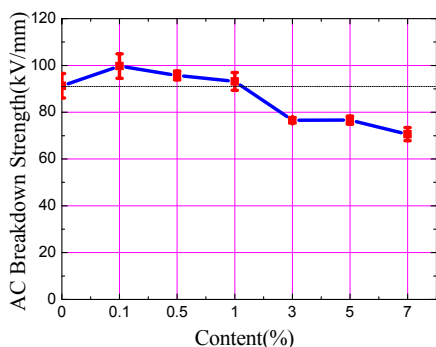


Figure 4 AC breakdown strength of LDPE/ZnO nanocomposites.

4.5 ELECTRICAL TREEING LIFE

Figure 5 shows the electrical treeing life of LDPE and LDPE/ZnO nanocomposites. The electrical treeing time of LDPE is less than 2 h, while that of the nanocomposites has tremendously increased. The 10 wt % nanocomposite has a life time more than 100 h, which is more than 50 times the value of LDPE.

As has been confirmed by many researches, electrical tree mainly propagate along the boundary of the large spherulites, which contain lots of large void defects and impurities [19]. Molecular arrangement in their boundary can readily introduce macroscopic voids and advance the tree growing by intensifying partial discharge. It has been stated afore that

nanoparticle fillers can reduce the formation of this kind of voids. Electrical tree grows and propagates mainly via partial discharge erosion at the sites of the macroscopic defects and the concentration of electric field [37]. Inorganic nanoparticles have much better discharge resistance than LDPE, so they can be a barrier for discharge development and thus can change the path of electrical treeing propagating, expending the energy for treeing. This favors the elongation of treeing life in nanocomposites. In addition, the large amounts of trap levels in the nanocomposite can depress the generation of high energy electrons, which can prohibit the corresponding chain-break reactions. ZnO can introduce much better treeing inhibition properties in LDPE than the general nanoparticles because of the following characteristics it has. Firstly, ZnO is a super light stabilizer and it can absorb the UV light energy released by partial discharge, reducing the polymer degradation reactions induced by UV radiation. Secondly, the precipitation of semiconducting ZnO on the surface of the tree channel can relieve the local electric field concentration and homogenize the electric field distribution, behaving like the high field seeker [38-40]. This effect can alleviate the partial discharge in the electric field intensified regions. So ZnO fillers can elongate the electrical treeing life of LDPE immensely.

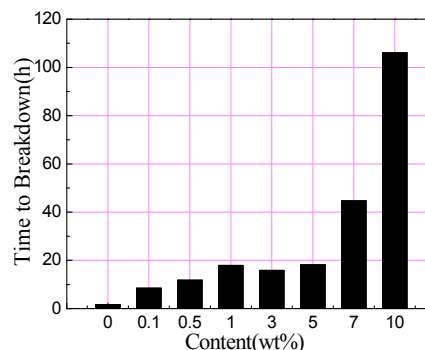


Figure 5 Electrical treeing life of LDPE/ZnO nanocomposites.

4.6 SPACE CHARGE DISTRIBUTION

The space charge distributions are showed in Figure 6. As has been clearly showed in Figure 6, for LDPE, immediately after the voltage is applied, both electrons and holes are injected from the cathode and anode separately. The charge distribution is symmetric. After 900 s, large amounts of heterocharges and homocharges are formed around the cathode, with a maximum density of about 6 C/m^3 .

Space charges in polymers can be divided into homocharges and heterocharges. Generally, homocharges are formed by electrons or holes injection from the cathode and anode. Heterocharges are from the ionization of impurities or decompositions of polymer itself. It is reasonable to assume impurities are derived mainly from fragments of polymerization catalyst, absorbed water, oxygen molecules, and additives introduced into the polymer during fabrication processes.

For LDPE, heterocharges come from the impurity ionization and ionized charge transport. If the electric field near the cathode is strong enough, then the impurities will be ionized, the positive ions accumulate near the cathode and the negative

electrons move toward the anode. Positive ions are saturated and accumulated around the cathode quickly since it is difficult for them to be neutralized via mass transport across the electrode interface, while both of the electrons by ionization and cathode injection become more and more and move toward the anode gradually. As heterocharges are formed near the cathode, the electric field near the cathode is enhanced significantly, which leads to a decrease of the potential barrier for electron injection. So the hole injection is generally covered.

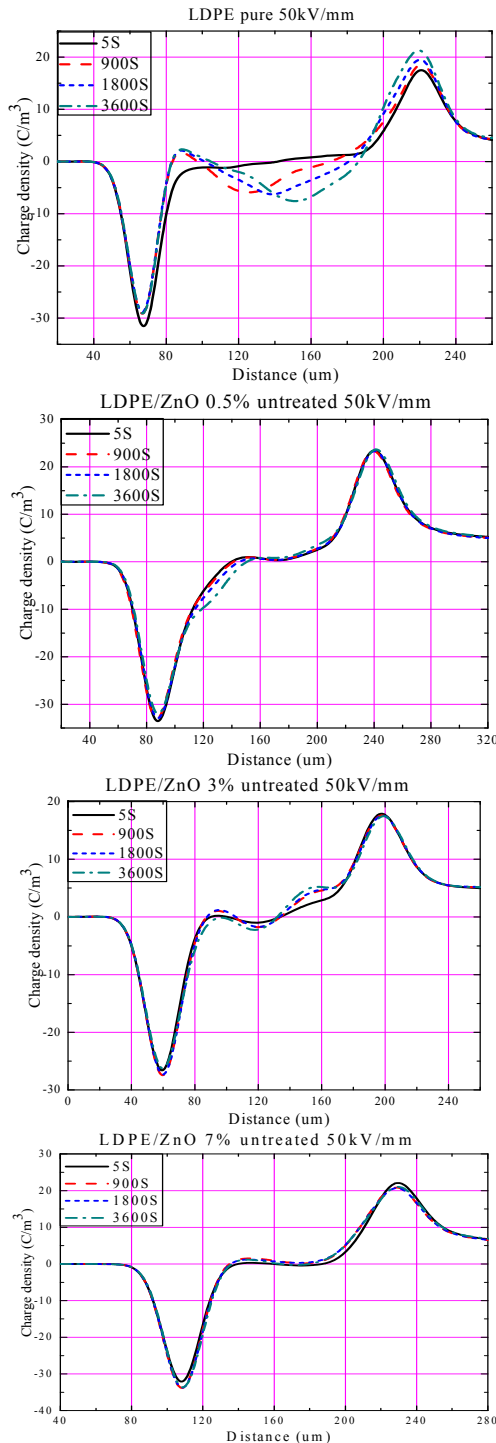


Figure 6 Space charge distributions in LDPE/ZnO nanocomposites under 50 kV/mm.

For the 0.5 wt% LDPE/ZnO nanocomposite, a small quantity of electrons and few holes are injected and homocharges are formed. The injected electrons and holes are limited close to the electrode and largely immobilized. No significant heterocharges are formed. Both the homocharges and heterocharges are largely decreased as compared with LDPE. For 3 wt% content, a little amount of heterocharges are gradually formed near the cathode. The few electrons behind the heterocharges may mainly come from ionization. Much more holes are injected from the anode. For 7 wt% content, we are surprised to find that hardly any homecharge by the electrode injection or ionized heterocharges are formed as compared with LDPE. It exhibits remarkable space charge suppression effect. The significant suppression of space charge in LDPE/ZnO nanocomposites is important for high voltage direct current cable development.

Generally, space charge accumulation in LDPE can be inhibited via increasing the electrical conduction^[41]. However, the conduction current is decreased significantly as showed before for 0.5wt% and 7wt% ZnO content. So the decreased space charge accumulation in 0.5wt% and 7wt% nanocomposites has been attributed to the introduction of deep trapping states(as showed by trap level distributions), which can largely enhance charge carrier trapping and reduce their mobility. The large amounts of deep trapping states (0.8-1 eV) are helpful to promote trapping of injected charge at the interface around the electrodes. The trapped charges near the injecting contact can create a thin homocharge layer and hence an internal field opposite to the applied field, reducing the effective field for charge injection from the contact and raising the injection potential barrier of electrons (or holes). The small amount of injected charge can be easily conducted out of the bulk. Therefore the homocharge accumulation is suppressed. This is consistent with the prediction made by Tanaka [22]. In addition, ions due to impurity ionization are limited to a much localized region as a result of the low charge mobility, i.e., the ionization is weakened. Thus hetero-space charges are restricted [26].

The mechanism about charge (electron for example) injection inhibition in the nanocomposite will be discussed as follows.

For an ohmic contact between the electrode and the polymer, an accumulation layer of electrons will be formed in the interface between the electrode and the polymer when the equilibrium is reached and the width of the electron layer can be expressed as follows [42]

$$\lambda = \frac{\pi}{2} \left(\frac{2kT\epsilon_r\epsilon_0}{e^2 N_t} \right)^{1/2} \exp\left(\frac{\psi_i - \chi - E_t}{2kT}\right) \quad (2)$$

Where k is the Boltzmann constant, T is temperature, ϵ_0 is the vacuum dielectric permittivity, ϵ_r is the relative permittivity of the polymer, e is the electron charge, N_t is the trap density, ψ_i is the work function of the polymer, χ is the electron affinity of the polymer. As can be easily found, the width of the electron accumulation layer decreases with the increase of trap density if the trap depth is unchanged.

According to the trap level distribution obtained before, the trap density of the nanocomposite can be 4 times that of LDPE at most. So the width of the electron accumulation should be less than 1/2 of that in LDPE.

The compact distribution of the accumulated electrons in the nanocomposite can significantly reduce the electric field around the interface. This can significantly reduce the electron injection current as showed by Schottky emission formula [43],

$$j_{in} = AT^2 \exp\left(-\frac{e\phi}{kT}\right) \exp\left(\frac{e}{kT} \sqrt{\frac{eE(0,t)}{4\pi\epsilon_r\epsilon_0}}\right) \quad (3)$$

Where j_{in} is the injection current density, A is constant, ϕ is the potential barrier for electron injection $E(0,t)$ is the interface electric field. If the electric field at the interface in the LDPE and the nanocomposite are 36 kV/mm and 9 kV/mm respectively, then the injection electron current density for the nanocomposite is about 1/20 of that for LDPE. The much less injection current will contribute to the space charge suppression in the LDPE/ZnO nanocomposites.

5 CONCLUSIONS

LDPE/ZnO nanocomposites are prepared by melt blending method. The trap level density, conduction current, electrical breakdown strength, electrical treeing life and space charge distributions are investigated. It's found that large amounts of traps are introduced by ZnO nanoparticle fillers. The conduction current is significantly decreased. The electrical breakdown strength is enhanced for lower ZnO contents. Electrical treeing life is significantly elongated. The study indicates that the improvements on the electrical properties, especially the charge transport and space charge suppression in the nanocomposites are closely related to the significant trap density increase. The effect of UV light absorption and electric field homogenization can also contribute to the enhancement of electrical breakdown strength and elongation of treeing life.

ACKNOWLEDGEMENT

This work was the key program supported by Program of National Key Basis and Development Plan (973) under Contract No. 2009CB724505 and National Natural Science Foundation program under Contract No. 60871073.

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